

Midterm exam (Nov. 2024)

Question 1: introduction to embedded systems

(12 marks)

a) True or false

1. Embedded systems always require a complex, high-power microcontroller. ()
2. A key benefit of embedded systems is the potential for lower costs due to the use of less expensive components and streamlined manufacturing. ()
3. A bike computer's primary sensor detects wheel movement using GPS. ()
4. Real-time constraints are more critical for a bike computer than for an engine control unit (ECU). ()
5. Engine Control Units (ECUs) primarily manage the entertainment system in a car. ()
6. Field-Programmable Gate Arrays (FPGAs) can be used to implement microprocessors within a System on a Chip (SoC). ()
7. A "soft core" processor is physically etched onto the silicon of an SoC. ()
8. The Internet of Things (IoT) involves distributed networks of embedded systems, often with sensors and controllers. ()

b) choose the best answer

1. What does IP stand for in the context of embedded systems?

- | | |
|--------------------------|------------------------|
| a) Instruction Protocol | b) Internet Protocol |
| c) Intellectual Property | d) Internal Peripheral |

2. Which of the following is NOT a type of IP Core?

- | | |
|---------------------|----------------------|
| a) Hard IP Core | b) Soft IP Core |
| c) Flexible IP Core | d) None of the above |

3. A Soft IP Core is typically delivered as:

- | | |
|---------------------------|---|
| a) A physical chip | b) An encrypted or unreadable source file |
| c) A compiled binary file | d) A set of documentation |

4. What is the role of EDA companies in the ARM ecosystem?

- | | |
|--|--------------------------------------|
| a) They design ARM processors. | b) They turn Soft IPs into Hard IPs. |
| c) They develop software for ARM processors. | d) They market ARM products. |

5. The Cortex-M series is designed for which type of applications?

- | | |
|--|--|
| a) High-performance applications with complex OS | b) Real-time applications requiring high reliability |
| c) Cost-sensitive microcontroller applications | d) Security-critical applications |

6. Which Cortex series is suitable for applications like engine control units (ECUs)?

- | | |
|-------------|--------------|
| a) Cortex-A | b) Cortex-R |
| c) Cortex-M | d) SecurCore |

7. Which of the following is an application of the Cortex-A series?

- | | |
|-------------------------------|---------------------------------|
| a) Smart sensors | b) Smartphones |
| c) Automotive braking systems | d) High-security access control |

8. ARM7 processors are:

- | | |
|-----------|-----------|
| a) 64-bit | b) 32-bit |
| c) 16-bit | d) 8-bit |

9. Which byte order stores the most significant byte first?

- | | |
|------------------|----------------------|
| a) Little Endian | b) Big Endian |
| c) Mixed Endian | d) None of the above |

10. Which of the following is an advantage of Little Endian?

- | | |
|--|------------------------------|
| a) Easier to print hexadecimal values | b) Easier to determine sign |
| c) Easier for multi-precision arithmetic | d) Easier to compare numbers |

12. What is the role of the SPSR?

- | | |
|---|---|
| a) Stores the status of the current program | b) Saves the CPSR value when entering exception modes |
| c) Points to the top of the stack | d) Stores the return address after a branch |

14. Which mode is entered on an ARM7 processor immediately after a reset?

- | | |
|-------------------|--------------------|
| a) User Mode | b) Supervisor Mode |
| c) Interrupt Mode | d) Abort Mode |

c) Match the following in pairs (one from column A with one from column B)

Column A	Column B	A→B
1.Hard IP Core	a. Scalable, delivered as source code (often encrypted)	
2.Soft IP Core	b. High-performance processors for applications with OS support	
3.Cortex-A Series	c. Pre-designed, physical block of logic; fixed electrical properties	
4.Cortex-M Series	d. Cost-sensitive, deterministic microcontroller applications	
5.ARM7TDMI's: the "D" sands for....	e. Hardware debugging	

Question 2: ARM assembly programming (8 marks)

a) Write ARM7 assembly program that finds the maximum value of int array of 100 items.

AREA array_max, CODE, READONLY

ENTRY

; Define array size and address

.....// Load address of array into R1

.....// Load array size (100) into R2

.....// Initialize max (R0) with the first element of the array

loop SUBS R2, R2, #1 ; Decrement counter, set flags

.....// Branch to end if counter is zero

.....// Load the next element from the array into R3 (use post indexing)

.....// Compare current max (R0) with current element (R3)

.....// If R0 >= R3, go to the next element in the array

.....// If R3 > R0, update max with R3

next B loop // Branch back to loop

end // Maximum value is now in R0

stop B stop // Infinite loop to halt execution

AREA data, DATA, READWRITE

array//Example initial values (first 10)

END

b) Create ARM7 Assembly program for a 2nd order difference equation without using past y values (FIR filter).

Difference equation: $y[n] = a_0 \cdot x[n] + a_1 \cdot x[n-1] + a_2 \cdot x[n-2]$

mov r0, #0x4 //r0=a0

mov r1, #0x2 //r1=a1

mov r2, #0x1 //r2=a2

// Memory buffers

x_buffer EQU 0x40000000 // start address of x[n]

y_buffer EQU 0x40000100 // start address of y[n]

buffer_size EQU 97 // no of output samples to be computed

start: //Initialize pointers and counter

.....// r3 points to start address of x[n]

.....// r4 points to start address of y[n]

MOV r5, #buffer_size // r5 = Loop counter

loop:

.....// Get current input x[n] into r6 (replace with actual input reading)

.....// $y[n] = a_0 \cdot x[n]$; Calculate r9=y[n]

// Access previous x samples (assuming 32-bit samples)

LDR r7, [r3, #-4] // r7 = x[n-1]

.....// $y[n] += a_1 \cdot x[n-1]$

.....// r8 = x[n-2]

.....// $y[n] += a_2 \cdot x[n-2]$

STR r9, [r4] // Store computed $y[n]=r9$ into its memory buffer

//Increment pointers and decrement counter

.....// increment x[n] buffer pointer

.....// increment y[n] buffer pointer

SUBS r5, r5, #1 // decrement no of output samples to be computed

BNE loop

stop: B stop // Infinite loop